# EQUIVARIANT GAMMA EXPANSION OF AUGMENTED MATROID CHOW RINGS

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*Editorial note.* Original author statement, entire paired proof, definitions and supporting arguments retained. No added mathematical hypothesis, proof repair or correctness certification.

*Editorial note.* Full article context is retained to preserve internal references, prerequisite constructions, and native diagrams. Only this task primary statement is counted.

*Editorial note.* Mechanical package adaptation: source line 19 loads shuffle.sty, but the article contains no shuffle command. Only this unused package-loading line is excluded; mathematical macros and body retained.

*Allowed standard prerequisites (editorial).*

- Standard graph, matroid and enumerative combinatorics. The source article definitions, cited standard results and bibliography are retained.. Original prerequisites; no new assumptions or author-proof replacements.

## 1. Introduction

Given a finite sequence, there are many features that one can ask whether the sequence possesses. A sequence $\{a_i\}_{i=0}^d$ or a polynomial $f(t) = \sum_{i=0}^d a_i t^i$ of degree $d$ is said to be *palindromic* if

$$a_j = a_{d-j} \quad \text{for all } j = 0, 1, \ldots, d.$$

It is said to be *unimodal* if there is some $0 \leq j \leq d$ such that

$$a_0 \leq a_1 \leq \cdots \leq a_j \geq \cdots \geq a_{d-1} \geq a_d.$$

A palindromic polynomial $f(t)$ can be expressed uniquely as

$$f(t) = \sum_{k=0}^{\lfloor \frac{d}{2} \rfloor} \gamma_k t^k (1+t)^{d-2k}.$$

The above expression is known as the $\gamma$-expansion of $f$ and $(\gamma_1, \gamma_2, \ldots, \gamma_{\lfloor \frac{d}{2} \rfloor})$ is called the $\gamma$-vector of $f$. We say the polynomial $f$ is *$\gamma$-positive* (or *$\gamma$-nonnegative*) if $\gamma_k \geq 0$ for all $k$. It is not hard to see that a polynomial being $\gamma$-positive implies that it is both palindromic and unimodal.

There are several reasons why the $\gamma$-positivity of a polynomial has attracted significant attention. One is the abundance of well-studied combinatorial sequences that are $\gamma$-positive; in many cases, their $\gamma$-coefficients admit interesting combinatorial interpretations, and sometimes this positivity phenomenon together with these interpretations can be lifted to a richer algebraic or representation-theoretic setting. A classical example is the $\gamma$-positivity of the Eulerian polynomial, first studied by Foata and Strehl [15], which was later extended to the level of symmetric functions [28, 29]. On the other hand, $\gamma$-positivity also appears naturally in geometric contexts. In particular, Gal conjectured that the $h$-polynomial of every flag simplicial sphere is $\gamma$-positive [16, Conjecture 2.1.7]. In this framework, the Eulerian polynomial arises as the $h$-polynomial of the dual permutahedron, which is itself a flag simplicial sphere. For more details about $\gamma$-positivity, we refer the readers to the survey [3] by Athanasiadis.

Let $M$ be a loopless matroid on the ground set $E$. The *Chow ring of the matroid $M$* encodes the structure of the lattice of flats $\mathcal{L}(M)$ and is defined by

$$A(M) \coloneqq \mathbb{R}[x_F : F \in \mathcal{L}(M) \setminus \{\emptyset\}]/(I + J)$$

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where $I = (x_F x_{F'} : F, F' \text{ not comparable})$ and $J = (\sum_{F: i \in x_F} x_F : i \in E)$. Similarly, the *augmented Chow ring of $M$* is defined by

$$\widetilde{A}(M) \coloneqq \mathbb{R}[\{x_F : F \in \mathcal{L}(M) \setminus \{[n]\}\} \cup \{y_i : i \in E\}]/(\widetilde{I} + \widetilde{J})$$

where $\widetilde{I} = (x_F x_{F'} : F, F' \text{ not comparable}) + (y_i x_F : i \notin F)$ and $\widetilde{J} = (y_i - \sum_{F: i \notin F} x_F : i \in E)$. The ordinary Chow ring $A(M)$ can be recovered from the augmented one by quotienting out the ideal generated by the variables $y_i$. Despite the seemingly complicated presentation, the Chow ring $A(M)$ has a nice $\mathbb{R}$-basis, which we refered to as the *Feichtner-Yuzvinsky basis*

$$\mathrm{FY}(M) \coloneqq \left\{ x_{F_1}^{a_1} x_{F_2}^{a_2} \cdots x_{F_\ell}^{a_\ell} : \begin{smallmatrix} \emptyset = F_0 \subsetneq F_1 \subsetneq F_2 \subsetneq \ldots \subsetneq F_\ell \text{ for } 0 \leq \ell \leq n \\ 1 \leq a_i \leq \mathrm{rk}_M(F_i) - \mathrm{rk}_M(F_{i-1}) - 1 \end{smallmatrix} \right\}.$$

It is the standard monomials obtained from the Gröbner basis of the ideal $I + J$ found by Feichtner and Yuzvinsky [11]. A similar Feichtner-Yuzvinsky basis for $\widetilde{A}(M)$,

$$\widetilde{\mathrm{FY}}(M) \coloneqq \left\{ x_{F_1}^{a_1} x_{F_2}^{a_2} \cdots x_{F_\ell}^{a_\ell} : \begin{smallmatrix} \emptyset \subsetneq F_1 \subsetneq F_2 \subsetneq \ldots \subsetneq F_\ell \text{ for } 0 \leq \ell \leq n \\ 1 \leq a_1 \leq \mathrm{rk}_M(F_1), \\ 1 \leq a_i \leq \mathrm{rk}_M(F_i) - \mathrm{rk}_M(F_{i-1}) - 1 \text{ for } i \geq 2 \end{smallmatrix} \right\},$$

was found by the author [22, 23] and independently by Chris Eur (see [27]). Using these bases, one can compute the Hilbert polynomials of the corresponding Chow rings and augmented Chow rings, which are now commonly referred to as the *Chow polynomial* and the *augmented Chow polynomials of a matroid*.

The Chow rings and augmented Chow rings of matroids play central roles in the resolution of the long-standing Rota-Welsh conjecture and Dowling-Wilson top-heavy conjecture, respectively. A key ingredient in both proofs is that these rings satisfy the so-called Kähler package (see [1] for Chow rings and [6] for augmented Chow rings). This package includes a Poincaré duality, which implies the Chow and augmented Chow polynomials are palindromic, and the Hard Lefschetz Theorem, which implies these polynomials are unimodal. The Chow and augmented Chow polynomials of a matroid were proved to be $\gamma$-positive by Ferroni, Matherne, Stevens, and Vecchi [12], and independently by Wang [12, p.33], using a recursive argument based on the semismall decomposition introduced in [7]. However, an explicit expression or a combinatorial interpretation of the $\gamma_k$ coefficients remains unknown. Moreover, these polynomials were conjectured to satisfy an even stronger positivity phenomenon. This was formulated by Huh and Stevens [35, Conjecture 4.3.3], and independently by Ferroni and Schröter [14, Conjecture 8.18] (for Chow polynomials) as follows.

**Conjecture 1.1.** *For every loopless matroid $M$, the Chow polynomial and the augmented Chow polynomial of $M$ have only real roots.*

Taking into account group actions on matroids, Angarone, Nathanson, and Reiner [2] extended several known and conjectural inequalities for Chow polynomials to the equivariant (representation-theoretic) and Burnside (permutation-representation) setting. In particular, they conjectured that the Chow ring $A(M)$, viewed as a graded $\mathbb{C}G$-modules, is equivariantly $\gamma$-positive (see Section 3 for the definition), and moreover equivariantly $PF_\infty$ [2, Conjecture 4.4 (ii)], which is an equivariant analog of Conjecture 1.1.

This paper is the full version of the extended abstract [25] presented at FPSAC 2025. We prove the equivariant $\gamma$-positivity conjecture of Angarone, Nathanson, and Reiner. Our approach further provides an explicit interpretation of the $\gamma$-coefficients, which is new even in the non-equivariant setting. We expect that this interpretation will provide further insight into Conjecture 1.1 and its equivariant analog.

In Section 3, we establish the equivariant $\gamma$-positivity conjecture. In Section 4, we apply our $\gamma$-positivity formulas to uniform matroids. From the perspective of matroid Chow rings, our results extend the combinatorial interpretations of the Charney-Davis quantity for uniform matroids studied by Hameister, Rao, and Simposon [18], as well as for general matroids by Liao [24]. From the perspective of symmetric functions and representations of symmetric groups, our results generalize the Schur-$\gamma$-positivity of the Eulerian and binomial Eulerian quasisymmetric functions studied by Shareshian and Wachs [28, 29], Linusson, Shareshian, and Wachs [26], and Athanasiadis [3]. In Section 5, we apply results from Section 4 to address a problem raised by Athanasiadis [3, Problem 2.5] concerning a $(p, q)$-analog of the $\gamma$-expansion of the binomial Eulerian polynomial. In Section 6, we comment on applications of our equivariant $\gamma$-positivity results on braid matroids. In Section 7, we discuss recent developments related to this work that have appeared since the preliminary version was posted on `arXiv`.

In the course of preparing this paper, we learned that Stump [36] independently obtained an interpretation of the (non-equivariant) $\gamma$-coefficients for Chow and augmented Chow polynomials of matroids. His approach uses a specialization of the *Poincaré extended* **ab**-*index* introduced in [9], expressing the $\gamma$-coefficient as the number of maximal chains in $\mathcal{L}(M)$ with prescribed descent set under an $R$-labeling. His results agree with ours, as a consequence of Stanley [31, Theorem 3.1]. In future work, we aim to further investigate the relationship between the Feichtner-Yuzvinsky basis and the Poincaré extended **ab**-index and study the equivariant analog of the extended **ab**-index.

## 2. Group actions on posets

Let $P$ be a finite poset with a unique minimal element $\hat{0}$ and a unique maximal element $\hat{1}$. If $P$ is graded of rank $n$ with rank function $\mathrm{rk}_P : P \longrightarrow \{0, 1, \ldots, n\}$, then for $S \subseteq [n-1]$, the *rank selected subposet* of $P$ is

$$P_S := \{x \in P \ : \ \mathrm{rk}_P(x) \in S\} \cup \{\hat{0}, \hat{1}\}.$$

Consider a group $G$ of automorphisms of $P$ acting on the poset $P$. The action of $G$ preserves the rank of the elements in $P$; hence for any $S \subseteq [n-1]$, the group $G$ permutes the maximal chains in $P_S$. Let us denote $\alpha_P(S)$ the permutation representation of $G$ generated by the maximal chains in $P_S$. Consider the virtual representation

$$\beta_P(S) := \sum_{T \subseteq S} (-1)^{|S|-|T|} \alpha_P(T),$$

which by Möbius inversion, is equivalent to

$$\alpha_P(S) = \sum_{T \subseteq S} \beta_P(T). \tag{1}$$

Here $\alpha_P(S)$ and $\beta_P(S)$ can be viewed as the equivariant analogs of the flag $f$-vector and flag $h$-vector of the poset $P$, respectively. See [34, Section 3.13] for more details.

**Theorem 2.1** (Stanley [32])**.** *If $P$ is Cohen-Macaulay, then $\beta_P(S)$ is a genuine representation of $G$ and*

$$\beta_P(S) \cong_G \tilde{H}_{|S|-1}(P_S).$$

Throughout this paper, the poset $P$ we care about is always the lattice of flats of a matroid. It is well known that the lattice of flats of a matroid is always Cohen-Macaulay.

## 3. Equivariant $\gamma$-positivity

Let $G$ be a finite group. For a $\mathbb{C}G$-module $M$, let $[M]$ be the isomorphism class of $\mathbb{C}G$-modules containing $M$.

**Definition 3.1.** The Grothendieck ring $R_{\mathbb{C}}(G)$ of $\mathbb{C}G$-modules is a free abelian group having the transversal of isomorphism classes of simple $\mathbb{C}G$-modules $\{[S_i], \ldots, [S_{cc(G)}]\}$ as a $\mathbb{Z}$-basis, where $cc(G)$ is the number of conjugacy classes of $G$, with the addition and multiplication relations given by

$$[S_i] + [S_j] := [S_i \oplus S_j] \quad \text{and} \quad [S_i] \cdot [S_j] := [S_i \otimes_{\mathbb{C}} S_j]$$

for $1 \leq i, j \leq cc(G)$ and extended linearly over $\mathbb{C}$. Now every element $A$ in $R_{\mathbb{C}}(G)$ has a unique expression as $a = \sum_{i=1}^{cc(G)} a_i[S_i]$. We say $A$ is a *genuine representation* of $G$ if $a_i \geq 0$ for all $i$, denoted by $A \geq_{R_{\mathbb{C}}(G)} 0$.

Given a graded $\mathbb{C}G$-module $V = \bigoplus_i V_i$, define the *equivariant Hilbert series* of $V$ to be the formal power series

$$\mathsf{Hilb}_G(V, t) = \sum_i [V_i] t^i \in R_{\mathbb{C}}(G)[[t]].$$

**Definition 3.2.** Let $V = \bigoplus_{i=0}^{d} V_i$ be a finite dimensional graded $\mathbb{C}G$-module. We say that $V$ is *(G-)equivariant $\gamma$-positive* if its equivariant Hilbert series can be expressed as

$$\mathsf{Hilb}_G(V, t) = \sum_{i=0}^{d} [V_i] t^i = \sum_{k=0}^{\lfloor \frac{d}{2} \rfloor} \gamma_k t^k (1+t)^{d-2k}$$

and the uniquely defined coefficient $\gamma_k \in R_{\mathbb{C}}(G)$ is a class of a genuine representation of $G$ over $\mathbb{C}$ for all $k$, i.e. $\gamma_k \geq_{R_{\mathbb{C}}(G)} 0$ for all $k$.

Let $M$ be a loopless matroid of rank $r$ on the ground set $E$ with the lattice of flats $\mathcal{L}(M)$. Write $A(M)_{\mathbb{C}} \coloneqq A(M) \otimes_{\mathbb{R}} \mathbb{C}$ and $\widetilde{A}(M)_{\mathbb{C}} \coloneqq \widetilde{A}(M) \otimes_{\mathbb{R}} \mathbb{C}$.

From now on, let $G$ be a group of automorphisms of the matroid $M$. The induced action of $G$ on the Chow ring $A(M)_{\mathbb{C}} = \bigoplus_{i=0}^{r-1} A^i_{\mathbb{C}}$ endows each graded piece $A^i_{\mathbb{C}}$ a $\mathbb{C}G$-module structure. The same holds for the augmented Chow ring $\widetilde{A}(M)_{\mathbb{C}} = \bigoplus_{i=0}^{r} \widetilde{A}^i_{\mathbb{C}}$. The following conjecture was proposed by Angarone, Nathanson, and Reiner.

**Conjecture 3.3** ([2, Conjecture 5.2])**.** *For any (loopless) matroid $M$ of rank $r$ and any group $G$ of automorphisms of $M$, the matroid Chow ring $A(M)_{\mathbb{C}} = \bigoplus_{i=0}^{r-1} A^i_{\mathbb{C}}$ is (G-)equivariant $\gamma$-positive.*

In the remainder of this section, we prove Conjecture 3.3 in Theorem 3.5, as well as its augmented counterpart in Theorem 3.9. Note that the Feichtner-Yuzvinsky bases $\mathrm{FY}(M)$ and $\widetilde{\mathrm{FY}}(M)$ form permutation bases of $A(M)_{\mathbb{C}}$ and $\widetilde{A}(M)_{\mathbb{C}}$, respectively, under the action of $G$.

**Proposition 3.4.** *Let $M$ be a loopless matroid of rank $r$.*

(i) *The $G$-equivariant Hilbert series of $A(M)_{\mathbb{C}}$ is given by*

$$\mathsf{Hilb}_G(A(M)_{\mathbb{C}}, t) = \sum_{S \subseteq [r-1]} \phi_{S,r}(t)[\alpha_{\mathcal{L}(M)}(S)],$$

*where for a subset $S = \{s_1 < s_2 < \cdots < s_\ell\} \subseteq [r-1]$,*

$$\phi_{S,r}(t) \coloneqq t^{|S|}[s_1 - 1]_t[s_2 - s_1 - 1]_t \cdots [s_\ell - s_{\ell-1} - 1]_t[r - s_\ell]_t.$$

*In particular, $\phi_{\emptyset,r}(t) = [r]_t$.*

(ii) *The $G$-equivariant Hilbert series of $\widetilde{A}(M)_{\mathbb{C}}$ is given by*

$$\mathsf{Hilb}_G(\widetilde{A}(M)_{\mathbb{C}}, t) = \sum_{S \subseteq [r-1]} \psi_{S,r}(t)[\alpha_{\mathcal{L}(M)}(S)],$$

*where for a subset $S = \{s_1 < s_2 < \cdots < s_\ell\} \subseteq [r-1]$,*

$$\psi_{S,r}(t) = \begin{cases} t^{|S|}[s_1]_t[s_2 - s_1 - 1]_t \ldots [s_\ell - s_{\ell-1} - 1]_t[r - s_\ell]_t & \text{if } S \neq \emptyset \\ [r+1]_t & \text{if } S = \emptyset \end{cases}.$$

*Proof.* For (i), recall that the Feichtenr-Yuzvinsky basis $\mathrm{FY}(M)$ consists of monomials $x_{F_1}^{a_1} x_{F_2}^{a_2} \ldots x_{F_\ell}^{a_\ell}$ indexed by chains (including the empty chain)

$$\emptyset \neq F_1 \subsetneq F_2 \subsetneq \cdots \subsetneq F_\ell \subseteq E$$

such that the exponent $a_j$ satisfies $1 \leq a_j \leq \mathrm{rk}(F_j) - \mathrm{rk}(F_{j-1}) - 1$ for $j = 1, 2, \ldots, \ell$, where we set $F_0 = \emptyset$. For any subset $S = \{s_1 < s_2 < \cdots < s_\ell\} \subseteq [r-1]$, the permutation $\mathbb{C}G$-module generated by chains in $\mathcal{L}(M)$ with rank set $S$ is

$$\begin{aligned} \alpha_{\mathcal{L}(M)}(S) &= \mathbb{C}G\{F_1 \subsetneq \cdots \subsetneq F_\ell : \mathrm{rk}(F_i) = s_i \ \forall i\}. \\ &\cong \mathbb{C}G\{F_1 \subsetneq \cdots \subsetneq F_\ell \subsetneq E : \mathrm{rk}(F_i) = s_i \ \forall i\}. \end{aligned}$$

Hence, as a graded $\mathbb{C}G$-module, $A(M)_{\mathbb{C}}$ decomposes as a direct sum of $\alpha_{\mathcal{L}(M)}(S)$ over all $S \subseteq [r-1]$.

Assume that $\mathsf{Hilb}_G(A(M)_{\mathbb{C}}, t) = \sum_{S \subseteq [r-1]} \phi_{S,r}(t)[\alpha_{\mathcal{L}(M)}(S)]$ for some polynomials $\phi_{S,r}(t)$. To determine $\phi_{S,t}(t)$, we consider a map $f : \mathrm{FY}(M) \longrightarrow 2^{[r-1]}$ defined by

$$f(x_{F_1}^{a_1} x_{F_2}^{a_2} \ldots x_{F_\ell}^{a_\ell} x_E^i) = \{\mathrm{rk}(F_1), \mathrm{rk}(F_2), \ldots, \mathrm{rk}(F_\ell)\}$$

where all flats $F_i$ are distinct and are not $E$. Since for any $S = \{s_1 < \cdots < s_\ell\} \subseteq [r-1]$, the inverse image of $f$ on $S$ consists of monomials

$$f^{-1}(S) = \left\{ x_{F_1}^{a_1} x_{F_2}^{a_2} \ldots x_{F_\ell}^{a_\ell} x_E^i : \begin{smallmatrix} \emptyset = F_0 \subsetneq F_1 \subsetneq \cdots \subsetneq F_\ell \subsetneq E \text{ with } \mathrm{rk}(F_j) = s_j \\ 1 \leq a_j \leq s_j - s_{j-1} - 1 \text{ for } j = 1, 2 \ldots, \ell \\ 0 \leq i \leq r - s_\ell - 1 \end{smallmatrix} \right\},$$

Therefore,

$$\begin{aligned} \phi_{S,r}(t) &= (t + t^2 + \cdots + t^{s_1 - 1}) \ldots (t + t^2 + \cdots + t^{s_\ell - s_{\ell-1} - 1})(1 + t + \cdots + t^{r - s_\ell - 1}) \\ &= t^{|S|}[s_1 - 1]_t[s_2 - s_1 - 1]_t \ldots [s_\ell - s_{\ell-1} - 1]_t[r - s_\ell]_t. \end{aligned}$$

In particular, when $S$ is an empty set, the inverse image $f^{-1}(\emptyset) = \{x_E^i \ : \ 0 \leq i \leq r-1\}$ and $\phi_{\emptyset,r}(t) = [r]_t$.

For (ii), the argument is analogous. The basis $\widetilde{\mathrm{FY}}(M)$ consists of monomials indexed by chains as above, with exponents satisfying $1 \le a_1 \le \mathrm{rk}(F_1)$ and $1 \le a_j \le \mathrm{rk}(F_j) - \mathrm{rk}(F_{j-1}) - 1$ for $j \ge 2$. Thus, we write

$$\mathsf{Hilb}_G(\widetilde{A}(M)_{\mathbb{C}}, t) = \sum_{S \subseteq [r-1]} \psi_{S,r}(t)\, [\alpha_{\mathcal{L}(M)}(S)].$$

Define $g : \widetilde{\mathrm{FY}}(M) \longrightarrow 2^{[r-1]}$ by

$$g(x_{F_1}^{a_1} \cdots x_{F_\ell}^{a_\ell} x_E^i) = \{\mathrm{rk}(F_1), \ldots, \mathrm{rk}(F_\ell)\}.$$

Then counting $g^{-1}(S)$ for $S = \{s_1 < \ldots < s_\ell\} \subseteq [r-1]$ gives

$$\psi_{S,r}(t) = \begin{cases} t^{|S|}[s_1]_t[s_2 - s_1 - 1]_t \cdots [s_\ell - s_{\ell-1} - 1]_t[r - s_\ell]_t & \text{if } S \neq \emptyset \\ [r+1]_t & \text{if } S = \emptyset \end{cases},$$

as claimed. □

The following theorem establishes Conjecture 1.1 and provides explicit expressions for the (equivariant) $\gamma$-coefficients of $A(M)$. It also generalizes the positivity of the (equivariant) Charney–Davis quantities for $A(M)$ obtained in [24, Lemma 4.9].

**Theorem 3.5.** *For any loopless matroid $M$ of rank $r$ on the ground set $E$ with the lattice of flats $\mathcal{L}(M)$, the Chow ring $A(M)$ is equivariant $\gamma$-positive with the following $\gamma$-expansion*

$$\begin{aligned} \mathsf{Hilb}_G(A(M)_{\mathbb{C}}, t) &= \sum_{S \in \mathrm{Stab}([2,r-1])} [\beta_{\mathcal{L}(M)}(S)] t^{|S|} (1+t)^{r-1-2|S|} \\ &= \sum_{k=0}^{\lfloor \frac{r-1}{2} \rfloor} \left( \sum_{\substack{S \in \mathrm{Stab}([2,r-1]) \\ |S|=k}} [\beta_{\mathcal{L}(M)}(S)] \right) t^k (1+t)^{r-1-2k}, \end{aligned}$$

*where $\beta_{\mathcal{L}(M)}(S) \cong \tilde{H}_{|S|-1}(\mathcal{L}(M)_S)$ as a $\mathbb{C}G$-module*

*Proof.* Notice that the polynomial $\phi_{S,r}(t)$ in Proposition 3.4 (i) is equal to 0 if $s_j - s_{j-1} = 1$ for some $1 \le j \le \ell$. Hence $\phi_{S,r}(t)$ takes nonzero value only when $S \in \mathrm{Stab}([2, r-1])$. The identity in Proposition 3.4 can be rephrased as follows:

$$\begin{aligned} \mathsf{Hilb}_G(A(M)_{\mathbb{C}}, t) &= \sum_{S \in \mathrm{Stab}([2,r-1])} \phi_{S,r}(t) [\alpha_{\mathcal{L}(M)}(S)] \\ &= \sum_{S \in \mathrm{Stab}([2,r-1])} \phi_{S,r}(t) \sum_{T : T \subseteq S} [\beta_{\mathcal{L}(M)}(T)] \\ &= \sum_{T \in \mathrm{Stab}([2,r-1])} \left( \sum_{T \subseteq S \subseteq [r-1]} \phi_{S,r}(t) \right) [\beta_{\mathcal{L}(M)}(T)], \end{aligned}$$

where the second equality follows from equation (1). It turns out that this complicated sum has a neat reformulation:

$$\sum_{T \subseteq S \subseteq [r-1]} \phi_{S,r}(t) = t^{|T|}(1+t)^{r-1-2|T|}.$$

This fact and its proof will be presented in Lemma 3.6. Combining it with Theorem 2.1, which states that $\beta_{\mathcal{L}(M)}(T)$ is isomorphic to $\tilde{H}_{|T|-1}(\mathcal{L}(M)_T)$ as $\mathbb{C}G$-modules, we obtain

$$\begin{aligned} \mathsf{Hilb}_G(A(M)_{\mathbb{C}}, t) &= \sum_{T \in \mathrm{Stab}([2,r-1])} [\beta_{\mathcal{L}(M)}(T)] t^{|T|} (1+t)^{r-1-2|T|} \\ &= \sum_{T \in \mathrm{Stab}([2,r-1])} [\tilde{H}_{|T|-1}(\mathcal{L}(M)_T)] t^{|T|} (1+t)^{r-1-2|T|}. \end{aligned}$$

□

*Remark* 3.1. The $\gamma$-expansion in Theorem 3.5, together with its augmented counterpart in Theorem 3.9, recovers and extends several known $\gamma$-expansions from [28], [29], and [3]. Thus, the present approach yields a unified framework for these results; see Section 4 for further details.

*Remark* 3.2. It is relatively rare for equivariant $\gamma$-positivity to hold in such generality. To the best of the author's knowledge, another instance of this phenomenon appears in the context of *Rees products of posets* by Athanasiadis [4]. In that setting, a key step in establishing equivariant $\gamma$-positivity is [4, Lemma 3.1], which involves a pair of identities closely resembling Lemma 3.6 and its augmented counterpart, Lemma 3.10. It would be of interest to understand whether there is a common underlying structure that explains this parallel between the two settings.

**Lemma 3.6.** *For a positive integer $n \geq 2$ and a subset $T \in \mathrm{Stab}([2, n-1])$,*

$$\sum_{T \subseteq S \subseteq [n-1]} \phi_{S,n}(t) = t^{|T|}(1+t)^{n-1-2|T|}, \tag{2}$$

*where $\phi_{S,n}(t) = 0$ if $S \notin \mathrm{Stab}([2, n-1])$ and for $S = \{s_1 < \cdots < s_\ell\} \in \mathrm{Stab}([2, n-1])$,*

$$\phi_{S,n}(t) = t^{|S|}[s_1 - 1]_t[s_2 - s_1 - 1]_t \dots [s_\ell - s_{\ell-1} - 1]_t[n - s_\ell]_t.$$

We give a combinatorial proof of Lemma 3.6. This is done by double-counting a combinatorial model constructed below. The model is motivated by Angarone, Nathanson, and Reiner's combinatorial proof of [2, Theorem 1.1].

**Definition 3.7.** For $n \geq 2$, let $\mathcal{W}_n$ be the set of sequences $\mathbf{w} = w_1 w_2 \dots w_{n-1}$ over the alphabet set $\{\bullet, \times, _(\text{a blank space})\}$ satsifying the following conditions:

(W1) If $w_i = \bullet$, then $w_{i-1} = \times$ (hence $w_1 \neq \bullet$ and there is no consecutive $\bullet$ in $\mathbf{w}$).

Let $S = \{s_1 < s_2 < \dots < s_\ell\} \in \mathrm{Stab}([2, n-1])$ be the set of indices $s$ with $w_s = \bullet$.

(W2) If there is $s_i < j < s_{i+1}$ such that $w_j = \times$, then $w_j = w_{j+1} = \dots = w_{s_{i+1}-1} = \times$.

(W3) If there is $s_\ell < j < n$ such that $w_j = \times$, then $w_j = w_{j+1} = \dots = w_{n-1} = \times$.

For $\mathbf{w} \in \mathcal{W}_n$, let $\mathsf{Dot}(\mathbf{w}) = \{i \in [n-1] : w_i = \bullet\}$ and $\mathsf{cro}(\mathbf{w})$ be the number of crossings $\times$ in $\mathbf{w}$.

*Proof of Lemma 3.6.* We prove (2) by double-counting

$$\sum_{\substack{\mathbf{w} \in \mathcal{W}_n \\ \mathsf{Dot}(\mathbf{w}) \supseteq T}} t^{\mathsf{cro}(\mathbf{w})}.$$

**First counting.** Fix a subset $S \in \mathrm{Stab}([2, n-1])$, say $S = \{s_1 < \dots < s_\ell\}$. For a sequence $\mathbf{w} = w_1 \dots w_{n-1} \in \mathcal{W}_n$ with $\mathsf{Dot}(\mathbf{w}) = S$, we have $w_{s_j} = \bullet$ for all $j$. By (W1), each $\bullet$ forces $w_{s_j - 1} = \times$. Between consecutive positions $s_{j-1}$ and $s_j$, the symbol at $s_j - 1$ is fixed to be $\times$, and we may choose any number of additional consecutive $\times$ immediately to its left, ranging from 0 to $s_j - s_{j-1} - 2$. This contributes a factor

$$t\left(1 + t + \cdots + t^{s_j - s_{j-1} - 2}\right) = t[s_j - s_{j-1} - 1]_t.$$

A similar argument applies to the initial segment before $s_1$ and the final segment after $s_\ell$, with the latter contributing $[n - s_\ell]_t$. It follows that, if $S$ is nonempty,

$$\sum_{\substack{\mathbf{w} \in \mathcal{W}_n \\ \mathsf{Dot}(\mathbf{w}) = S}} t^{\mathsf{cro}(\mathbf{w})} = t^{|S|}[s_1 - 1]_t[s_2 - s_1 - 1]_t \cdots [s_\ell - s_{\ell-1} - 1]_t[n - s_\ell]_t = \phi_{S,n}(t);$$

While $S = \emptyset$, $\sum_{\substack{\mathbf{w} \in \mathcal{W}_n \\ \mathsf{Dot}(\mathbf{w}) = \emptyset}} t^{\mathsf{cro}(\mathbf{w})} = 1 + t + \cdots + t^{n-1} = [n]_t = \phi_{\emptyset,n}(t)$. Summing over all $S \supseteq T$ gives

$$\sum_{\substack{\mathbf{w} \in \mathcal{W}_n \\ \mathsf{Dot}(\mathbf{w}) \supseteq T}} t^{\mathsf{cro}(\mathbf{w})} = \sum_{T \subseteq S \subseteq [n-1]} \phi_{S,n}(t).$$

**Second counting.** Fix $\mathbf{w} \in \mathcal{W}_n$ with $\mathsf{Dot}(\mathbf{w}) \supseteq T$. For each $i \in T$, we have $w_i = \bullet$, and hence by (W1), $w_{i-1} = \times$. These forced $\times$ contribute a factor $t^{|T|}$. The remaining positions, namely those not of the form $i$ or $i-1$ for $i \in T$, form a set of size $n-1-2|T|$. On these positions, we may independently choose whether to place a $\times$ or leave a blank, contributing a factor $(1+t)^{n-1-2|T|}$. Given such a choice, the conditions (W2) and (W3) uniquely determine the placement of the remaining $\bullet$ symbols so that $\mathbf{w} \in \mathcal{W}_n$. Therefore,

$$\sum_{\substack{\mathbf{w} \in \mathcal{W}_n \\ \mathsf{Dot}(\mathbf{w}) \supseteq T}} t^{\mathsf{cro}(\mathbf{w})} = t^{|T|}(1+t)^{n-1-2|T|}.$$

Comparing the two expressions completes the proof. □

In the following, we illustrate the procedure in the proof of Lemma 3.6 by some examples.

**Example 3.8.** Let $n = 12$ and $T = \{3, 9\} \in \mathrm{Stab}([2, 11])$.

**First counting.** Fix $S \supseteq T$.

*Case 1:* If $S = \{3, 5, 9\}$, we place $\bullet$ at positions $3, 5, 9$, and by (W1) we must place $\times$ at $2, 4, 8$. The remaining positions are initially blank:

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
|---|---|---|---|---|---|---|---|---|---|---|
| _ | × | • | × | • | _ | _ | × | • | _ | _ |

Between consecutive $\bullet$'s, we may extend the forced $\times$'s to the left. For instance, between $w_5 = \bullet$ and $w_9 = \bullet$, we may choose

$$(_, _, \times), \quad (_, \times, \times), \quad (\times, \times, \times),$$

contributing $t + t^2 + t^3 = t[3]_t$. Proceeding similarly for all segments, we obtain

$$\sum_{\substack{\mathbf{w} \in \mathcal{W}_{12} \\ \mathsf{Dot}(\mathbf{w}) = \{3,5,9\}}} t^{\mathsf{cro}(\mathbf{w})} = t[2]_t \cdot t[1]_t \cdot t[3]_t \cdot [3]_t = t^3 [2]_t [1]_t [3]_t [3]_t.$$

*Case 2:* If $S = \{3, 9, 11\}$, we place $\bullet$ at $3, 9, 11$ and $\times$ at $2, 8, 10$:

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
|---|---|---|---|---|---|---|---|---|---|---|
| _ | × | • | _ | _ | _ | _ | × | • | × | • |

Since $11 = n - 1 \in S$, there is no trailing segment after the last $\bullet$. The contributions from the segments give

$$\sum_{\substack{\mathbf{w} \in \mathcal{W}_{12} \\ \mathsf{Dot}(\mathbf{w}) = \{3,9,11\}}} t^{\mathsf{cro}(\mathbf{w})} = t[2]_t \cdot t[5]_t \cdot t[1]_t = t^3 [2]_t [5]_t [1]_t.$$

**Second counting.** We count sequences $\mathbf{w}$ with $\mathsf{Dot}(\mathbf{w}) \supseteq T = \{3, 9\}$. Then $w_3 = w_9 = \bullet$ and $w_2 = w_8 = \times$ are forced. The remaining positions

$$\{1, 4, 5, 6, 7, 10, 11\}$$

can be chosen independently to be either $\times$ or blank, contributing

$$t^{|\{2,8\}|}(1 + t)^{|\{1,4,5,6,7,10,11\}|} = t^2 (1 + t)^{12-1-2\cdot 2}.$$

For example, if we choose $w_1 = w_4 = w_5 = w_{10} = \times$, we obtain

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
|---|---|---|---|---|---|---|---|---|---|---|
| × | × | • | × | × | _ | _ | × | • | × | _ |

.

There is then a unique way to insert additional $\bullet$'s so that $\mathbf{w} \in \mathcal{W}_{12}$, namely:

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
|---|---|---|---|---|---|---|---|---|---|---|
| × | × | • | × | × | • | _ | × | • | × | • |

.

This illustrates the identity

$$\sum_{\substack{\mathbf{w} \in \mathcal{W}_{12} \\ \mathsf{Dot}(\mathbf{w}) \supseteq T}} t^{\mathsf{cro}(\mathbf{w})} = t^2 (1 + t)^7.$$

The following theorem is the augmented counterpart of Theorem 3.5. It generalizes the positivity of the (equivariant) Charney–Davis quantities for the augmented Chow ring $\widetilde{A}(M)$ obtained in [24, Lemma 4.12]. The proof follows the same strategy as in the Chow case, with a modified combinatorial model.

**Theorem 3.9.** *For any loopless matroid $M$ of rank $r$ on the ground set $E$ with the lattice of flats $\mathcal{L}(M)$, the augmented Chow ring $\widetilde{A}(M)_{\mathbb{C}}$ is equivariant $\gamma$-positive with the following $\gamma$-expansion*

$$\begin{aligned} \mathsf{Hilb}_G(\widetilde{A}(M)_{\mathbb{C}}, t) &= \sum_{S \in \mathrm{Stab}([r-1])} [\beta_{\mathcal{L}(M)}(S)] t^{|S|} (1 + t)^{r - 2|S|} \\ &= \sum_{k=0}^{\lfloor \frac{r}{2} \rfloor} \left( \sum_{\substack{S \in \mathrm{Stab}([r-1]) \\ |S| = k}} [\beta_{\mathcal{L}(M)}(S)] \right) t^k (1 + t)^{r-2k}. \end{aligned}$$

*where $\beta_{\mathcal{L}(M)}(S) \cong \tilde{H}_{|S|-1}(\mathcal{L}(M)_S)$ as $\mathbb{C}G$-modules.*

*Proof.* The argument is parallel to that of Theorem 3.5. By Proposition 3.4 (ii), we have

$$\mathsf{Hilb}_G(\widetilde{A}(M)_{\mathbb{C}}, t) = \sum_{S \subseteq [r-1]} \psi_{S,r}(t)\, [\alpha_{\mathcal{L}(M)}(S)].$$

Since $\psi_{S,r}(t) = 0$ whenever $S$ contains consecutive integers, the sum reduces to $S \in \mathrm{Stab}([r-1])$. Thus,

$$\begin{aligned}\mathsf{Hilb}_G(\widetilde{A}(M)_{\mathbb{C}}, t) &= \sum_{S \in \mathrm{Stab}([r-1])} \psi_{S,r}(t)[\alpha_{\mathcal{L}(M)}(S)] \\ &= \sum_{S \in \mathrm{Stab}([r-1])} \psi_{S,r}(t) \sum_{T \subseteq S} [\beta_{\mathcal{L}(M)}(T)] \\ &= \sum_{T \in \mathrm{Stab}([r-1])} \left( \sum_{T \subseteq S \subseteq [r-1]} \psi_{S,r}(t) \right) [\beta_{\mathcal{L}(M)}(T)].\end{aligned}$$

By Lemma 3.10 below, the inner sum satisfies

$$\sum_{T \subseteq S \subseteq [r-1]} \psi_{S,r}(t) = t^{|T|}(1+t)^{r-2|T|},$$

Combining this with Theorem 2.1, we obtain

$$\begin{aligned}\mathsf{Hilb}_G(\widetilde{A}(M)_{\mathbb{C}}, t) &= \sum_{T \in \mathrm{Stab}([r-1])} [\beta_{\mathcal{L}(M)}(T)] t^{|T|}(1+t)^{r-2|T|} \\ &= \sum_{T \in \mathrm{Stab}([r-1])} [\tilde{H}_{|T|-1}(\mathcal{L}(M)_T)] t^{|T|}(1+t)^{r-2|T|}.\end{aligned}$$

□

**Lemma 3.10.** *For $n \geq 2$ and subset $T \in \mathrm{Stab}([n-1])$,*

$$\sum_{T \subseteq S \subseteq [n-1]} \psi_{S,n}(t) = t^{|T|}(1+t)^{n-2|T|} \tag{3}$$

*where $\psi_{S,n}(t) = 0$ if $S \notin \mathrm{Stab}([n-1])$ and for $S = \{s_1 < \cdots < s_\ell\} \in \mathrm{Stab}([n-1])$,*

$$\psi_{S,n}(t) = \begin{cases} t^{|S|}[s_1]_t[s_2 - s_1 - 1]_t \dots [s_\ell - s_{\ell-1} - 1]_t[n - s_\ell]_t & \text{if } S \neq \emptyset \\ [n+1]_t & \text{if } S = \emptyset \end{cases}.$$

We use another combinatorial model slightly different from $\mathcal{W}_n$ in the non-augmented case to prove Lemma 3.10.

**Definition 3.11.** For $n \geq 2$, let $\widetilde{\mathcal{W}}_n$ be the set of sequences $\mathbf{w} = w_0 w_1 w_2 \dots w_{n-1}$ of length $n$ over the alphabet set $\{\bullet, \times, _(\text{a blank space})\}$ satisfying conditions (W1), (W2), and (W3) from Definition 3.7.

Condition (W1) implies that $w_0 \neq \bullet$ and no two consecutive entries of $\mathbf{w}$ are both $\bullet$. As in the non-augmented case, we define $\mathsf{Dot}(\mathbf{w})$ to be the set of positions occupied by $\bullet$, and $\mathsf{cro}(\mathbf{w})$ to be the number of $\times$'s in $\mathbf{w}$ for $\mathbf{w} \in \widetilde{\mathcal{W}}_n$.

*Proof of Lemma 3.10.* The argument is parallel to that of Lemma 3.6. We compute

$$\sum_{\substack{\mathbf{w} \in \widetilde{\mathcal{W}}_n \\ \mathsf{Dot}(\mathbf{w}) \supseteq T}} t^{\mathsf{cro}(\mathbf{w})}$$

in two ways.

**First counting.** Fix $S = \{s_1 < \cdots < s_\ell\} \in \mathrm{Stab}([n-1])$ with $S \supseteq T$. As in the non-augmented case, the structure of $\mathbf{w} \in \widetilde{\mathcal{W}}_n$ with $\mathsf{Dot}(\mathbf{w}) = S$ yields

$$\sum_{\substack{\mathbf{w} \in \widetilde{\mathcal{W}}_n \\ \mathsf{Dot}(\mathbf{w}) = S}} t^{\mathsf{cro}(\mathbf{w})} = \psi_{S,n}(t),$$

where the contribution from each interval $[s_{j-1}, s_j]$ is given by $[s_j - s_{j-1} - 1]_t$ except when $j = 1$ it is $[s_1]_t$, and the final segment contributes either $[n - s_\ell]_t$ or is absorbed into the special case $S = \emptyset$. Summing over all $S \supseteq T$ gives

$$\sum_{\substack{\mathbf{w}\in\widetilde{\mathcal{W}}_n \\ \mathsf{Dot}(\mathbf{w})\supseteq T}} t^{\mathsf{cro}(\mathbf{w})} = \sum_{T\subseteq S\subseteq[n-1]} \psi_{S,n}(t).$$

**Second counting.** We instead construct sequences with $\mathsf{Dot}(\mathbf{w}) \supseteq T$ directly. First, for each $i \in T$, we force the pattern $w_i = \bullet$ and $w_{i-1} = \times$, contributing a factor $t^{|T|}$. After removing these forced positions, there remain $n - 2|T|$ free positions. In each such position, we independently choose either a $\times$ or a blank space, contributing a factor $(1+t)^{n-2|T|}$. Given any such choice, there is a unique way to insert additional $\times$'s so that the resulting sequence lies in $\widetilde{\mathcal{W}}_n$. Hence,

$$\sum_{\substack{\mathbf{w}\in\widetilde{\mathcal{W}}_n \\ \mathsf{Dot}(\mathbf{w})\supseteq T}} t^{\mathsf{cro}(\mathbf{w})} = t^{|T|}(1+t)^{n-2|T|}.$$

Equating the two expressions completes the proof. □

*Remark* 3.3. In a forthcoming paper, Ferroni, Matherne, and Vecchi [13] generalize the Chow and augmented Chow polynomials of matroids (geometric lattice) to a more general class of posets. They prove these polynomials are $\gamma$-positive for Cohen–Macaulay posets, using a strategy analogous to that of the present paper, though without the combinatorial proofs of Lemma 3.6 and Lemma 3.10 (See Section 7).

## 4. Uniform and $q$-uniform matroids

In this section, we consider the special case where $M$ is the uniform matroid $U_{r,n}$ of rank $r$ and $G = \mathfrak{S}_n$ is the symmetric group. In this setting, the lattice of flats can be identified with a rank-selected subposet of the Boolean lattice $B_n = (2^{[n]}, \subseteq)$. The equivariant $\gamma$-coefficients $\tilde{H}_{|R|-1}((B_n)_R)$ carry a well-studied $\mathbb{C}\mathfrak{S}_n$-module structure for any $R \subseteq [n-1]$. As a consequence, our results recover and extend several known $\gamma$-positivity phenomena in symmetric functions and permutation enumeration.

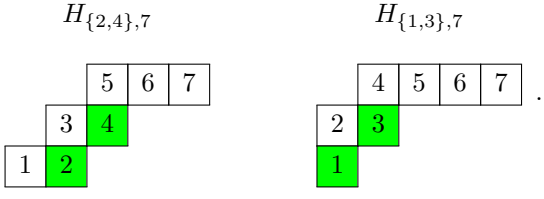
For background on representations of $\mathfrak{S}_n$, standard Young tableaux, and symmetric functions, we refer to Wachs [39] and Stanley [33]. For permutation statistics such as $\mathsf{exc}$, $\mathsf{inv}$, and $\mathsf{maj}$, see Chapter 1 of Stanley [34].

Let $\mathrm{ch}$ denote the *Frobenius characteristic map*, which assigns to each isomorphism class of $\mathbb{C}\mathfrak{S}_n$-modules a symmetric function of degree $n$. Let $\mathrm{ps}_q$ denote the *stable principal specialization*, defined by substituting $x_i = q^{i-1}$ in a symmetric function. In place of the equivariant Hilbert series, it is often convenient to use the graded Frobenius series to encode a graded $\mathfrak{S}_n$-representation as a polynomial with coefficients in symmetric functions. For a graded $\mathbb{C}\mathfrak{S}_n$-module $V = \bigoplus_i V_i$, the *graded Frobenius series* of V is defined by

$$\mathsf{grFrob}(V, t) := \sum_i \mathrm{ch}(V_i)\, t^i.$$

Let $\lambda$, $\mu$ be two partitions such that $\mu \subseteq \lambda$ (i.e. $\mu_i \leq \lambda_i$ for all $i$). The connected skew shape $\lambda/\mu$ is said to be a *ribbon* (or a *skew hook* or a *border strip*) if two consecutive rows always overlap in exactly one cell. A ribbon $H_{R,n}$ can be described in terms of the number $n$ of its cells and its *descent set* $R \subseteq [n-1]$. Given a ribbon of $n$ cells, we label its cells with numbers 1 through $n$, starting from the southwestern end cell, through the adjacent cell, ending at the northeastern end cell. We say a cell $i$ is a *descent* of the ribbon if cell $i+1$ is directly above cell $i$. Then the collection of all its descents is called its *descent set*.

**Example 4.1.** Let $n = 7$. Then the diagrams of $H_{\{2,4\},7}$ and $H_{\{1,3\},7}$ are



Let $S^{\lambda/\mu}$ be the *Specht module* of skew shape $\lambda/\mu$. If $\lambda/\mu$ is a ribbon $H_{R,n}$, the corresponding Specht module $S^{H_{R,n}}$ is known as a *Foulkes representation*. We summarize some facts of this representation from [32, Theorem 4.3] and [39, Theorem 3.4.4] in Theorem 4.2 (Note that Theorem 4.2 (i) was first proved by Solomon [30] in a different expression for any finite Coxeter group).

For a partition $\lambda$ of $n$, denoted by $\lambda \vdash n$, let $SYT(\lambda)$ be the set of standard Young tableaux of shape $\lambda$. For $P \in SYT(\lambda)$, let $\mathsf{DES}(P)$ be the set of entries (called descents) $i \in [n-1]$ such that $i+1$ appears in a lower row than $i$ in $P$, and $\mathsf{des}(P) = |\mathsf{DES}(P)|$.

**Theorem 4.2.** *Let $R \subseteq [n-1]$.*

(i) *Then*
$$\tilde{H}_{|R|-1}((B_n)_R) \cong_{\mathfrak{S}_n} S^{H_{R,n}} \cong_{\mathfrak{S}_n} \bigoplus_{\lambda \vdash n} b_\lambda(R) S^\lambda,$$
*where $b_\lambda(R)$ is the number of $P \in SYT(\lambda)$ for which $\mathsf{DES}(P) = R$.*

(ii) *Let $\mathbb{P}$ denote the set of positive integers and $\mathbb{P}_n$ the set of words of length $n$ over $\mathbb{P}$. For $w = w_1 \dots w_n \in \mathbb{P}_n$, we say $i \in \mathsf{DES}(w)$ if $w_i > w_{i+1}$ and write $\mathbf{x}_w = x_{w_1} \dots x_{w_n}$. Then the Frobenius characteristic is the ribbon Schur function*
$$\mathrm{ch}(\tilde{H}_{|R|-1}((B_n)_R)) = s_{H_{R,n}} = \sum_{\substack{w \in \mathbb{P}_n \\ \mathsf{DES}(w)=R}} \mathbf{x}_w,$$

Applying Theorem 3.5 to the uniform matroids $U_{r,n}$ together with Theorem 2.1, one obtains the graded Frobenius series of $A(U_{r,n})_{\mathbb{C}}$. Applying the operator $\prod_{i=1}^{n}(1-q^i)\mathrm{ps}_q(-)$ to the graded Frobenius series gives the Hilbert series of $q$-Uniform matroid $U_{r,n}(q)$ (See Section 3 in [24] for the explanation). Note that the stable principal specialization on the ribbon Schur function gives
$$\prod_{i=1}^{n}(1-q^i)\mathrm{ps}_q(s_{H_{R,n}}) = \sum_{\substack{\sigma \in \mathfrak{S}_n \\ \mathsf{DES}(\sigma)=R}} q^{\mathsf{maj}(\sigma^{-1})} = \sum_{\substack{\sigma \in \mathfrak{S}_n \\ \mathsf{DES}(\sigma)=R}} q^{\mathsf{inv}(\sigma)}$$
(see [29, Thm. 4.4] and its proof for a detailed discussion of the above identity). We obtain the following corollary, which generalizes Corollary 4.10 and 4.11 of [24]

**Corollary 4.3.** *For any positive integer $n$ and $1 \le r \le n$,*
$$\begin{aligned}\mathsf{grFrob}(A(U_{r,n})_{\mathbb{C}}, t) &= \sum_{R \in \mathrm{Stab}([2,r-1])} s_{H_{R,n}} t^{|R|}(1+t)^{r-1-2|R|} \\ &= \sum_{k=0}^{\lfloor \frac{r-1}{2} \rfloor} \left( \sum_{\substack{R \in \mathrm{Stab}([2,r-1]) \\ |R|=k}} s_{H_{R,n}} \right) t^k (1+t)^{r-1-2k}\end{aligned}$$
*and*
$$\mathsf{Hilb}(A(U_{r,n}(q)), t) = \sum_{k=0}^{\lfloor \frac{r-1}{2} \rfloor} \xi_{r,n,k}(q) t^k (1+t)^{r-1-2k}$$
*where*
$$\xi_{r,n,k}(q) = \sum_{\sigma} q^{\mathsf{maj}(\sigma^{-1})} = \sum_{\sigma} q^{\mathsf{inv}(\sigma)}$$
*and the sum runs through all $\sigma \in \mathfrak{S}_n$ for which $\mathsf{DES}(\sigma) \in \mathrm{Stab}([2, r-1])$ has $k$ elements.*

Applying Theorem 3.9 and the stable principal specialization as in Corollary 4.3, we obtain the augmented counterpart of Corollary 4.3, which generalizes Corollary 4.13 and 4.14 of [24].

**Corollary 4.4.** *For any positive integer $n$ and $1 \le r \le n$,*
$$\begin{aligned}\mathsf{grFrob}(\widetilde{A}(U_{r,n})_{\mathbb{C}}, t) &= \sum_{R \in \mathrm{Stab}([r-1])} s_{H_{R,n}} t^{|R|}(1+t)^{r-2|R|} \\ &= \sum_{k=0}^{\lfloor \frac{r}{2} \rfloor} \left( \sum_{\substack{R \in \mathrm{Stab}([r-1]) \\ |R|=k}} s_{H_{R,n}} \right) t^k (1+t)^{r-2k}\end{aligned}$$

*and*

$$\mathsf{Hilb}(\widetilde{A}(U_{r,n}(q)),t)=\sum_{k=0}^{\lfloor\frac{r}{2}\rfloor}\widetilde{\xi}_{r,n,k}(q)t^k(1+t)^{r-2k},$$

*where*

$$\widetilde{\xi}_{r,n,k}(q)=\sum_{\sigma}q^{\mathsf{maj}(\sigma^{-1})}=\sum_{\sigma}q^{\mathsf{inv}(\sigma)}$$

*and the sum runs through all* $\sigma\in\mathfrak{S}_n$ *for which* $\mathsf{DES}(\sigma)\in\mathrm{Stab}([r-1])$ *has* $k$ *elements.*

Recall from Corollary 4.2 and 4.4 of [24] that

$$\mathsf{grFrob}(A(U_{n-1,n}),t)=t^{-1}Q_n^0(\mathbf{x},t), \tag{4}$$

$$\mathsf{grFrob}(A(U_{n,n}),t)=\mathsf{grFrob}(\widetilde{A}(U_{n-1,n}),t)=Q_n(\mathbf{x},t), \tag{5}$$

$$\mathsf{grFrob}(\widetilde{A}(U_{n,n}),t)=\widetilde{Q}_n(\mathbf{x},t). \tag{6}$$

Here $Q_n^0$, $Q_n$, $\widetilde{Q}_n$ are the *Eulerian quasisymmetric functions* introduced by Shareshian and Wachs [28, 29]. They admit combinatorial interpretations as enumerators over the sets of derangements $D_n$ (permutations with no fixed points), permutations $\mathfrak{S}_n$, and *decorated permutations* $\widetilde{\mathfrak{S}}_n$ (permutations with two types of fixed points), respectively, satisfying

$$D_n\subset\mathfrak{S}_n\subset\widetilde{\mathfrak{S}}_n.$$

More precisely,

$$Q_n^0(\mathbf{x},t)=\sum_{\sigma\in D_n}F_{\mathsf{DEX}(\sigma),n}(\mathbf{x})t^{\mathsf{exc}(\sigma)},\quad Q_n(\mathbf{x},t)=\sum_{\sigma\in\mathfrak{S}_n}F_{\mathsf{DEX}(\sigma),n}(\mathbf{x})t^{\mathsf{exc}(\sigma)}, \tag{7}$$

and

$$\widetilde{Q}_n(\mathbf{x},t)=\sum_{\sigma\in\widetilde{\mathfrak{S}}_n}F_{\mathsf{DEX}(\sigma),n}(\mathbf{x})t^{\mathsf{exc}(\sigma)+1}. \tag{8}$$

Here $F_{S,n}(\mathbf{x})$ for $S\subseteq[n-1]$ denotes Gessel's *fundamental quasisymmetric function* [1]

$$F_{S,n}(\mathbf{x}):=\sum_{\substack{i_1\geq i_2\geq\cdots\geq i_n\\ j\in S\Rightarrow i_j>i_{j+1}}}x_{i_1}x_{i_2}\dots x_{i_n}\quad\text{with } F_{\phi,0}:=1.$$

See [24] for the definition of the descent-like statistic $\mathsf{DEX}$ on $\widetilde{\mathfrak{S}}_n$ and further details on these symmetric functions. Finally, by Lemma 6.20 of [23], applying the operator $\prod_{i=1}^n(1-q^i)\mathrm{ps}_q(-)$ to $Q_n^0$, $Q_n$, and $\widetilde{Q}_n$ yields the $q$-analogs of Eulerian type polynomials:

$$d_n(q,t)=\sum_{\sigma\in D_n}q^{\mathsf{maj}(\sigma)-\mathsf{exc}(\sigma)}t^{\mathsf{exc}(\sigma)},\quad A_n(q,t)=\sum_{\sigma\in\mathfrak{S}_n}q^{\mathsf{maj}(\sigma)-\mathsf{exc}(\sigma)}t^{\mathsf{exc}(\sigma)},$$

and

$$\widetilde{A}_n(q,t):=1+\sum_{k=1}^n\begin{bmatrix}n\\k\end{bmatrix}_q A_k(q,t)=\sum_{\sigma\in\widetilde{\mathfrak{S}}_n}q^{\mathsf{maj}(\sigma)-\mathsf{exc}(\sigma)}t^{\mathsf{exc}(\sigma)+1}.$$

Let $NDD_n$ be the set of words $w\in\mathbb{P}_n$ with no double descents, i.e. $\mathsf{DES}(w)\in\mathrm{Stab}([n-1])$. Combining (4) and Corollary 4.3, we recover the following collection of results from [28, Theorem 7.3, Equation (7.9)] and [29, Theorem 6.1].

**Corollary 4.5.** *For positive integer* $n\geq 2$*, we have*

$$\begin{aligned}Q_n^0(\mathbf{x},t)&=\sum_{R\in\mathrm{Stab}([2,n-2])}s_{H_{R,n}}t^{|R|}(1+t)^{n-1-2|R|}\\&=\sum_{\substack{w\in NDD_n\\ w_1\leq w_2,\ w_{n-1}\leq w_n}}\mathbf{x}_w t^{\mathsf{des}(w)+1}(1+t)^{n-2-2\mathsf{des}(w)}\\&=\sum_{k=0}^{\lfloor\frac{n-2}{2}\rfloor}\xi_{n,k}^0(\mathbf{x})t^{k+1}(1+t)^{n-2-2k},\end{aligned}$$

[1] Note that our definition of the fundamental quasisymmetric function is the reverse of the usual one. The usual definition is summed over $i_1\leq i_2\leq\cdots\leq i_n$ with $j\in S$ implying $i_j<i_{j+1}$.

*where*

$$\xi^0_{n,k}(\mathbf{x}) = \sum_{\substack{R\in \mathrm{Stab}([2,n-2])\\ |R|=k}} s_{H_{R,n}} = \sum_{\substack{w\in NDD_n\\ w_1\le w_2,\ w_{n-1}\le w_n\\ \mathsf{des}(w)=k}} \mathbf{x}_w.$$

*And*

$$d_n(q,t) = \sum_{k=0}^{\lfloor \frac{n-2}{2}\rfloor} \xi^0_{n,k}(q) t^{k+1}(1+t)^{n-2-2k},$$

*where*

$$\xi^0_{n,k}(q) = \sum_{\sigma} q^{\mathsf{maj}(\sigma^{-1})} = \sum_{\sigma} q^{\mathsf{inv}(\sigma)}$$

*and both of the sums run through $\sigma \in D_n$ which has $k$ descents and no double descents, no first, no last descent.*

Combining (5) with Corollary 4.3 and Corollary 4.4, we recover the following collection of results from [28, Theorem 7.3, Equation (7.8)] and [26, Equations (1.4) and (6.1)].

**Corollary 4.6.** *For positive integer $n \ge 1$, we have the following expressions*

$$\begin{aligned} Q_n(\mathbf{x},t) &= \sum_{R\in \mathrm{Stab}([2,n-1])} s_{H_{R,n}} t^{|R|}(1+t)^{n-1-2|R|} = \sum_{R\in \mathrm{Stab}([n-2])} s_{H_{R,n}} t^{|R|}(1+t)^{n-1-2|R|} \\ &= \sum_{\substack{w\in NDD_n\\ w_1\le w_2}} \mathbf{x}_w t^{\mathsf{des}(w)}(1+t)^{n-1-2\mathsf{des}(w)} = \sum_{\substack{w\in NDD_n\\ w_{n-1}\le w_n}} \mathbf{x}_w t^{\mathsf{des}(w)}(1+t)^{n-1-2\mathsf{des}(w)} \\ &= \sum_{k=0}^{\lfloor \frac{n-1}{2}\rfloor} \xi_{n,k}(\mathbf{x}) t^k (1+t)^{n-1-2k}, \end{aligned}$$

*where*

$$\begin{aligned} \xi_{n,k}(\mathbf{x}) &= \sum_{\substack{R\in \mathrm{Stab}([2,n-1])\\ |R|=k}} s_{H_{R,n}} = \sum_{\substack{R\in \mathrm{Stab}([n-2])\\ |R|=k}} s_{H_{R,n}} \\ &= \sum_{\substack{w\in NDD_n\\ w_1\le w_2\\ \mathsf{des}(w)=k}} \mathbf{x}_w = \sum_{\substack{w\in NDD_n\\ w_{n-1}\le w_n\\ \mathsf{des}(w)=k}} \mathbf{x}_w. \end{aligned}$$

*And*

$$A_n(q,t) = \sum_{k=0}^{\lfloor \frac{n-1}{2}\rfloor} \xi_{n,k}(q) t^{k+1}(1+t)^{n-1-2k},$$

*where*

$$\xi_{n,k}(q) = \sum_{\sigma} q^{\mathsf{maj}(\sigma^{-1})} = \sum_{\sigma} q^{\mathsf{inv}(\sigma)}$$

*and both of the sums run through either:*

- *$\sigma \in \mathfrak{S}_n$ with $k$ descents and no double descents, no first descent.*
- *$\sigma \in \mathfrak{S}_n$ with $k$ descents and no double descents, no last descent.*

*Remark* 4.1. In [3, Theorem 2.1], Athanasiadis summarizes different interpretations of the $\gamma$-coefficients for the Eulerian polynomial. In particular, the interpretations are given by counting permutations in $\mathfrak{S}_n$ related to the descent set in $\mathrm{Stab}([2,n-1])$ or in $\mathrm{Stab}([n-2])$. This fact is lifted to the $q$-analog and the symmetric function analog in Corollary 4.6. The agreement of the two interpretations regarding $\mathrm{Stab}([2,n-1])$ and $\mathrm{Stab}([n-2])$ was shown algebraically by (5).

Combining (6) and Corollary 4.4, we recover gives the following collection of results of [29, Theorems 3.4 and 4.5]

**Corollary 4.7** (cf. Shareshian and Wachs [29], Athanasiadis [3]). *For positive integer $n \geq 1$, we have*

$$\begin{aligned}\widetilde{Q}_n(\mathbf{x},t) &= \sum_{R\in \mathrm{Stab}([n-1])} s_{H_{R,n}} t^{|R|}(1+t)^{n-2|R|} \\ &= \sum_{w\in NDD_n} \mathbf{x}_w t^{\mathsf{des}(w)}(1+t)^{n-2\mathsf{des}(w)} \\ &= \sum_{k=0}^{\lfloor \frac{n}{2} \rfloor} \widetilde{\xi}_{n,k}(\mathbf{x}) t^k (1+t)^{n-2k},\end{aligned}$$

*where*

$$\widetilde{\xi}_{n,k}(\mathbf{x}) = \sum_{\substack{R\in \mathrm{Stab}([n-1]) \\ |R|=k}} s_{H_{R,n}} = \sum_{\substack{w\in NDD_n \\ \mathsf{des}(w)=k}} \mathbf{x}_w.$$

*And*

$$\widetilde{A}_n(q,t) = \sum_{k=0}^{\lfloor \frac{n}{2} \rfloor} \widetilde{\xi}_{n,k}(q) t^k (1+t)^{n-2k},$$

*where*

$$\widetilde{\xi}_{n,k}(q) = \sum_{\sigma} q^{\mathsf{maj}(\sigma^{-1})} = \sum_{\sigma} q^{\mathsf{inv}(\sigma)}$$

*and both of the sums run through $\sigma \in \mathfrak{S}_n$ with $k$ descents and no double descents.*

## 5. A problem proposed by Athanasiadis

In this section, we apply some results from Section 4 to address a problem proposed by Athanasiadis [3, Problem 2.5].

For $\sigma \in \mathfrak{S}_n$, we set

$$\mathsf{des}^*(\sigma) = \begin{cases} \mathsf{des}(\sigma) & \text{if } \sigma_1 = 1 \\ \mathsf{des}(\sigma) - 1 & \text{if } \sigma_1 > 1 \end{cases}.$$

Recall the following $(p,q)$-analog of the $\gamma$-expansion of the Eulerian polynomial in [3, Theorem 2.3].

**Theorem 5.1.** *For all positive integer $n \geq 1$,*

$$\sum_{\sigma\in\mathfrak{S}_n} p^{\mathsf{des}^*(\sigma)} q^{\mathsf{maj}(\sigma)-\mathsf{exc}(\sigma)} t^{\mathsf{exc}(\sigma)} = \sum_{k=0}^{\lfloor \frac{n-1}{2} \rfloor} \xi_{n,k}(p,q) t^k (1+t)^{n-1-2k}$$

*where*

$$\xi_{n,k}(p,q) = \sum_{\sigma} p^{\mathsf{des}(\sigma^{-1})} q^{\mathsf{maj}(\sigma^{-1})}$$

*summing over permutations $\sigma \in \mathfrak{S}_n$ with $k$ descents and no double descents, no last descent.*

Recall a similar $q$-analog of the $\gamma$-expansion of the binomial Eulerian polynomial $\widetilde{A}_n(t)$ found by Shareshian and Wachs [29] is given in Corollary 4.7. Athanasiadis proposed the following problem.

**Problem 5.2** ([3, Problem 2.5]). *Find a $p$-analog of the $q$-polynomial in Corollary 4.7 that is similar to Theorem 5.1.*

The idea of finding such a $(p,q)$-analog of $\widetilde{A}_n(t)$ is similar to the proof of Theorem 5.1. But the key step is the $\mathfrak{S}_n$-equivariant enumerator of decorated permutations in (8)[2].

Before answering Athanasiadis' question, let us first generalize Corollary 2.41 in [3], which states the $\gamma$-positivity for the coefficient of Schur-expansions of $Q_n^0(\mathbf{x},t)$ and $Q_n(\mathbf{x},t)$. Note that $Q_n$ and $Q_n^0$ are related to the graded Frobenius series of the Chow ring of $U_{n,n}$ and $U_{n-1,n}$ respectively. Here we generalize these results to all uniform matroids $U_{r,n}$ and their augmented counterparts.

Let

$$\mathsf{grFrob}(A(U_{r,n})_{\mathbb{C}},t) = \sum_{\lambda\vdash n} P_\lambda^r(t) s_\lambda \text{ and } \mathsf{grFrob}(\widetilde{A}(U_{r,n})_{\mathbb{C}},t) = \sum_{\lambda\vdash n} \widetilde{P_\lambda^r}(t) s_\lambda.$$

[2]This interpretation is proved in Liao [23, Theorem 6.21]. The original definition given by Shareshian and Wachs [29] is $\widetilde{Q}_n(\mathbf{x},t) := h_n + t\sum_{k=1}^n h_{n-k} Q_k(\mathbf{x},t)$.

**Corollary 5.3.** *For $\lambda \vdash n$, we have*

$$P_\lambda^r(t) = \sum_{\substack{P\in SYT(\lambda)\\ \mathsf{DES}(P)\in \mathrm{Stab}([2,r-1])}} t^{\mathsf{des}(P)}(1+t)^{r-1-2\mathsf{des}(P)} = \sum_{k=0}^{\lfloor \frac{r-1}{2}\rfloor} \xi_{r,\lambda,k}\ t^k(1+t)^{r-1-2k}$$

*and*

$$\widetilde{P}_\lambda^r(t) = \sum_{\substack{P\in SYT(\lambda)\\ \mathsf{DES}(P)\in \mathrm{Stab}([r-1])}} t^{\mathsf{des}(P)}(1+t)^{r-2\mathsf{des}(P)} = \sum_{k=0}^{\lfloor \frac{r}{2}\rfloor} \widetilde{\xi}_{r,\lambda,k}\ t^k(1+t)^{r-2k}$$

*where $\xi_{r,\lambda,k}$ (respectively, $\widetilde{\xi}_{r,\lambda,k}$) is the number of tableaux $P \in SYT(\lambda)$ for which $\mathsf{DES}(P) \in \mathrm{Stab}([2, r-1])$ (respectively, $\mathrm{Stab}([r-1])$) has $k$ elements. In particular, $P_\lambda^r(t)$ and $\widetilde{P}_\lambda^r(t)$ are palindromic and unimodal for all $\lambda$ and $r$.*

*Proof.* Consider the non-augmented case. Apply Corollary 4.3 and use Theorem 2.1 (i) to expand $s_{H_{R,n}}$ into Schur functions of regular shapes. We have

$$\mathsf{grFrob}(A(U_{r,n})_{\mathbb{C}}, t) = \sum_{\lambda\vdash n}\left(\sum_{R\in \mathrm{Stab}([2,r-1])} b_\lambda(R)t^{|R|}(1+t)^{n-2|R|}\right)s_\lambda(\mathbf{x}).$$

Comparing the coefficient of $s_\lambda(\mathbf{x})$, we obtain

$$P_\lambda^r(t) = \sum_{R\in \mathrm{Stab}([2,r-1])} b_\lambda(R)t^{|R|}(1+t)^{r-1-2|R|} = \sum_{\substack{P\in SYT(\lambda)\\ \mathsf{DES}(P)\in \mathrm{Stab}([2,r-1])}} t^{\mathsf{des}(P)}(1+t)^{r-1-2\mathsf{des}(P)}.$$

The augmented can be worked out similarly. □

*Remark* 5.1. For general $1 \le r \le n$, the symmetric function polynomials $\mathsf{grFrob}(A(U_{r,n})_{\mathbb{C}}, t)$ and $\mathsf{grFrob}(\widetilde{A}(U_{r,n})_{\mathbb{C}}, t)$ each has two difference expressions as shown in [24, Theorem 4.1 and Theorem 4.3]. Unfortunately, we do not have nice combinatorial expressions as the cases $r = n-1,\ n$ in (7) and (8).

In particular, when $r = n$ in the augmented case of Corollary 5.3, it states that the Schur-expansion of the binomial Eulerian quasisymmetric function $\widetilde{Q}_n(\mathbf{x}, t)$ has $\gamma$-positive coefficients. This will be used in the proof of Theorem 5.4.

We used stable principal specialization to obtain many interesting $q$-analogs in the previous sections. To obtain a $(p, q)$-analog of $\widetilde{A}_n(t)$, we need a different specialization. Given a formal power series $f(\mathbf{x}) = f(x_1, x_2, \ldots)$ and a positive integer $m \ge 1$, the *principal specialization* $\mathrm{ps}_{q,m}(f)$ of $f$ is obtain from $f$ by setting

$$x_i = \begin{cases} q^{i-1} & \text{if } 1 \le i \le m \\ 0 & \text{if } i \ge m+1 \end{cases}.$$

Then the principal specialization of $F_{S,n}$ (see [17, Lemma 5.2]) and $s_\lambda(\mathbf{x})$ (see [33, Proposition 7.19.12]) are given by the formulas

$$\sum_{m\ge 1} \mathrm{ps}_{q,m}(F_{S,n}(\mathbf{x}))p^{m-1} = \frac{p^{|S|}q^{\sum_{i\in S} i}}{(1-p)(1-pq)\ldots(1-pq^n)} \tag{9}$$

and

$$\sum_{m\ge 1} \mathrm{ps}_{q,m}(s_\lambda(\mathbf{x}))p^{m-1} = \frac{\sum_{P\in SYT(\lambda)} p^{\mathsf{des}(P)}q^{\mathsf{maj}(P)}}{(1-p)(1-pq)\ldots(1-pq^n)}. \tag{10}$$

For any decorated permutation $\sigma \in \widetilde{\mathfrak{S}}_n$, we set

$$\widetilde{\mathsf{des}}(\sigma) = \begin{cases} \mathsf{des}(\sigma) & \text{if } \sigma_1 = 0 \text{ or } 1 \\ \mathsf{des}(\sigma) - 1 & \text{if } \sigma_1 > 1 \end{cases}.$$

Now we are ready to state our $(p, q)$-analog for $\widetilde{A}_n(t)$.

**Theorem 5.4.** *For a positive integer $n \geq 1$,*

$$\sum_{\sigma\in\widetilde{\mathfrak{S}}_n} p^{\widetilde{\mathsf{des}}(\sigma)}q^{\mathsf{maj}(\sigma)-\mathsf{exc}(\sigma)}t^{\mathsf{exc}(\sigma)+1} = \sum_{\substack{\sigma\in\mathfrak{S}_n\\ \mathsf{DES}(\sigma)\in\mathrm{Stab}([n-1])}} p^{\mathsf{des}(\sigma^{-1})}q^{\mathsf{maj}(\sigma^{-1})}t^{\mathsf{des}(\sigma)}(1+t)^{n-2\mathsf{des}(\sigma)}$$
$$= \sum_{k=0}^{\lfloor \frac{n}{2} \rfloor} \widetilde{\xi}_{n,k}(p,q)t^k(1+t)^{n-2k}$$

*where*

$$\widetilde{\xi}_{n,k}(p,q) = \sum_{\sigma} p^{\mathsf{des}(\sigma^{-1})}q^{\mathsf{maj}(\sigma^{-1})}$$

*and the sum runs through $\sigma \in \mathfrak{S}_n$ with $k$ descents and no double descents.*

*Proof.* First applying principal specialization on (8) and by (9), we have

$$\sum_{m\geq 1} \mathrm{ps}_{q,m}(\widetilde{Q}_n(\mathbf{x},t))p^{m-1}$$
$$= \sum_{\sigma\in\widetilde{\mathfrak{S}}_n}\sum_{m\geq 1} \mathrm{ps}_{q,m}(F_{\mathsf{DEX}(\sigma),n}(\mathbf{x}))p^{m-1}t^{\mathsf{exc}(\sigma)+1}$$
$$= \frac{\sum_{\sigma\in\widetilde{\mathfrak{S}}_n} p^{\widetilde{\mathsf{des}}(\sigma)}q^{\mathsf{maj}(\sigma)-\mathsf{exc}(\sigma)}t^{\mathsf{exc}(\sigma)+1}}{(1-p)(1-pq)\dots(1-pq^n)} \quad \text{(by Lemma 6.20 in [23])}.$$

On the other hand,

$$\sum_{m\geq 1} \mathrm{ps}_{q,m}(\widetilde{Q}_n(\mathbf{x},t))p^{m-1}$$
$$= \sum_{\lambda\vdash n} \widetilde{P}^n_\lambda(t)\sum_{m\geq 1} \mathrm{ps}_{q,m}(s_\lambda(\mathbf{x}))p^{m-1}$$
$$= \frac{\sum_{\lambda\vdash n} \widetilde{P}^n_\lambda(t)\sum_{P\in SYT(\lambda)} p^{\mathsf{des}(P)}q^{\mathsf{maj}(P)}}{(1-p)(1-pq)\dots(1-pq^n)} \quad \text{(by (10))}.$$

Therefore, we have

$$\sum_{\sigma\in\widetilde{\mathfrak{S}}_n} p^{\widetilde{\mathsf{des}}(\sigma)}q^{\mathsf{maj}(\sigma)-\mathsf{exc}(\sigma)}t^{\mathsf{exc}(\sigma)+1} = \sum_{\lambda\vdash n}\sum_{P\in SYT(\lambda)} \widetilde{P}^n_\lambda(t)p^{\mathsf{des}(P)}q^{\mathsf{maj}(P)}$$
$$= \sum_{\lambda\vdash n}\sum_{P\in SYT(\lambda)}\sum_{\substack{Q\in SYT(\lambda)\\ \mathsf{DES}(Q)\in\mathrm{Stab}([n-1])}} p^{\mathsf{des}(P)}q^{\mathsf{maj}(P)}t^{\mathsf{des}(Q)}(1+t)^{n-2\mathsf{des}(Q)} \quad \text{(by Corollary 5.3)}$$
$$= \sum_{\substack{\sigma\in\mathfrak{S}_n\\ \mathsf{DES}(\sigma)\in\mathrm{Stab}([n-1])}} p^{\mathsf{des}(\sigma^{-1})}q^{\mathsf{maj}(\sigma^{-1})}t^{\mathsf{des}(\sigma)}(1+t)^{n-2\mathsf{des}(\sigma)}.$$

The last equality is obtained by applying the RSK algorithm described in Section 7.11 of Stanley [33, Lemma 7.23.1]. □

## 6. A remark on braid maroids

The *Braid matroids* $K_n$ are the matroid associated with the braid arrangement $\{x_i - x_j = 0\}_{1\leq i<j\leq n}$, and equivalently, the graphical matroid of the complete graph of $n$ vertices. Its lattice of flats is the intersection lattice of the braid arrangement, and is given by the partition lattice $\Pi_n$. In [12], Ferroni, Matherne, Stevens, and Vecchi proposed a problem of studying the Hilbert series of the Chow and augmented Chow ring of $K_n$. Applying Theorem 3.5 and Theorem 3.9, we have

$$\mathsf{Hilb}_G(A(K_n),t) = \sum_{S\in\mathrm{Stab}([2,n-2])} [\tilde{H}_{|S|-1}((\Pi_n)_S)]t^{|S|}(1+t)^{n-2-2|S|}$$

and

$$\mathsf{Hilb}_G(\widetilde{A}(K_n),t) = \sum_{S\in\mathrm{Stab}([n-2])} [\tilde{H}_{|S|-1}((\Pi_n)_S)]t^{|S|}(1+t)^{n-1-2|S|}.$$

These give an explicit connection between the (augmented) Chow ring of $K_n$ and the top reduced homology of the rank-selected partition lattice $(\Pi_n)_S$. The study of $\mathfrak{S}_n$-representation $\tilde{H}_{|S|-1}((\Pi_n)_S)$

is initiated from Stanley's classical paper [32], and has been studied in depth since then. A nice survey for this is Sundaram [38]. However, the $\mathfrak{S}_n$-module structures of $\tilde{H}_{|S|-1}((\Pi_n)_S)$ seem to vary a lot, and there is no unified characterization for every subset $S$ so far. Regarding the dimension (rank) of $\tilde{H}_{|S|-1}((\Pi_n)_S)$, using the theory of EL-labeling, there is a general formula for all subsets $S$, see Sundaram [37, Prop. 2.18] or Björner [5, p.276, Exercise 7.33].

## 7. New developments

Since the preliminary version of this paper was posted on `arXiv`, there have been many interesting developments.

7.1. **Chow polynomials for posets.** Ferroni, Matherne, and Vecchi [13] generalized the notion of Chow polynomials from matroids to locally finite, weakly ranked posets, placing them within the framework of Kazhdan–Lusztig–Stanley theory for posets.

Given such a poset $P$ and a choice of a distinguished element $\kappa$ in the incidence algebra of $P$, called a *$P$-kernel*, they define the corresponding *$\kappa$-Chow polynomial of $P$*. For suitable choices of $\kappa$, this construction recovers and extends Chow-type invariants arising in the study of Eulerian posets and the Kazhdan–Lusztig theory of Coxeter groups, and allows for a systematic investigation of their positivity phenomena.

In particular, when $\kappa$ is the characteristic polynomial of $P$ and $P$ is a lattice of flats of a matroid $M$, one recovers the Chow polynomial of $M$. As noted in Remark 3.3, using an approach similar to ours, they proved that if $P$ is Cohen–Macaulay, then both the characteristic Chow and augmented Chow polynomials are $\gamma$-positive [13, Theorem 4.25]. We may extend Theorems 3.5 and 3.9 to Cohen-Macaulay posets as well.

**Definition 7.1.** Let $P$ be a finite graded bounded poset of rank $r$, and $G$ be a group of automorphisms of $P$. Define the *(G-)equivariant Chow polynomial* and *(G-)equivariant augmented Chow polynomial* of $P$ [3], respectively, to be the following polynomials with coefficients in $R_{\mathbb{C}}(G)$:

$$\mathrm{H}_P(t) \coloneqq \sum_{S\subseteq[r-1]} \phi_{S,r}(t)[\alpha_P(S)]$$

and

$$^{\mathrm{aug}}\mathrm{H}_P(t) \coloneqq \sum_{S\subseteq[r-1]} \psi_{S,r}(t)[\alpha_P(S)],$$

where $\phi_{S,r}(t)$ and $\psi_{S,r}(t)$ are as in Proposition 3.4.

When $G$ is trivial (or by taking dimensions of $\alpha_P(S)$), following from [13, Theorem 4.1] and its left augmented counterpart, the above polynomials specialize to the non-equivariant Chow and augmented Chow polynomials of $P$. The same arguments as in Theorems 3.5 and 3.9 then yield the following extension.

**Theorem 7.2.** *Let $P$ be a finite bounded graded poset of rank $r$. Then:*

(i) *the equivariant Chow polynomial of $P$ admits the $\gamma$-expansion*

$$\mathrm{H}_P(t) = \sum_{S\in\mathrm{Stab}([2,r-1])} [\beta_P(S)]t^{|S|}(1+t)^{r-1-2|S|}, \text{ and}$$

(ii) *the equivariant augmented Chow polynomial of $P$ admits $\gamma$-expansion*

$$^{\mathrm{aug}}\mathrm{H}_P(t) = \sum_{S\in\mathrm{Stab}([r-1])} [\beta_P(S)]t^{|S|}(1+t)^{r-2|S|}.$$

*In particular, if $P$ is a Cohen-Macaulay poset, then $\mathrm{H}_P(t)$ and $^{\mathrm{aug}}\mathrm{H}_P(t)$ are equivariant $\gamma$-positive.*

Along these lines, Ferroni, Matherne, and Vecchi further extend Conjecture 1.1 to this borader setting.

**Conjecture 7.3** (Conjecture 4.26 in [13])**.** *Let $P$ be a Cohen–Macaulay poset. The characteristic Chow polynomial of $P$ is real-rooted.*

Note that the real-rootedness of the characteristic (left) augmented Chow polynomial of a Cohen–Macaulay poset would follow from [13, Corollary 4.6] together with the above conjecture.

[3] Recursive definitions of $\mathrm{H}_P(t)$ and $^{\mathrm{aug}}\mathrm{H}_P(t)$ analogous to the non-equivariant construction in [12] are given by Matherne and Nepel (personal communication).

7.2. **Uniform matroids $U_{r,n}$ and their generalizations.** The coefficients of the Chow and augmented Chow polynomials of the Boolean matroid $U_{n,n}$ are counted, respectively, by the numbers of loopless Schubert matroids and Schubert matroids of the corresponding ranks. This follows from the bijection described in [10, proof of Theorem 7.7] between the Feichtner–Yuzvinsky basis $\widetilde{\mathrm{FY}}(U_{n,n})$ and Schubert matroids. Hoster [19] extended this combinatorial interpretation to Chow and augmented Chow polynomials of uniform matroids $U_{r,n}$ for all $r$.

Brändén and Vecchi [8] proved that the Chow polynomials of uniform matroids $U_{r,n}$ are real-rooted, and also provided a new proof of the real-rootedness of the augmented Chow polynomials of $U_{r,n}$, independent of the earlier argument in [12, Theorem 1.10].

Hoster and Stump [20] used the non-equivariant analogs of Theorems 3.5 and 3.9 to prove real-rootedness of Chow and augmented Chow polynomials for posets $\widehat{P} = P \cup \{\hat{1}\}$ where $P$ is a simplicial poset with positive $h$-vector. In particular, their argument yields yet another proof of real-rootedness for uniform matroids.

7.3. **Braid matroids $K_n$.** Kannan and Kühne [21] obtained a closed formula for the generating function of the $\mathfrak{S}_n$-equivariant Chow polynomial $\mathrm{H}_{K_n}(t)$ of the braid matroid $K_n$, expressed in terms of the $\mathfrak{S}_n$-equivariant compactly supported Poincaré polynomial of $\mathcal{M}_{0,n+1}$. They further showed that the Chow polynomial of $K_n$ agrees with the Hilbert series of the Chow ring of the moduli space of genus-zero relative stable maps to $\mathbb{P}^1$.

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